Abstract

Molybdenum Trioxide (MoO_3) is a transition metal interesting in its potential as a material for surface doping diamond electronics. The material is thought to be hygroscopic which could inhibit its ability to donate acceptors to the diamond by reducing the concentration of acceptors available for surface transfer doping. This would reduce the performance of fabricated diamond devices. For this reason it was set out to investigate the process by which MoO_3 is affected by exposure to the environment. It was found that there is a definite mass increase of the films when exposed to the environment. This is a certain sign of absorption and adsorption at work, showing that the material is in fact hygroscopic. Thus if MoO_3 is to be utilised as a surface dopant for diamond semiconductors in the future a method of limiting the processes by which moisture is absorbed into the film when exposed to the environment must be developed.

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1. Introduction:

Diamond semi-conductors are an emerging technology. Single crystal Chemical Vapour deposition (CVD) diamond has been shown to possess a number of qualities making it a great choice for development for electronics; highest mobility of both holes and electrons in any wide band gap semiconductor, $4500cm^2V^{-1}s^{-1}$ for holes and $3800cm^2V^{-1}s^{-1}$ for electrons, providing great conductivity [1]. Highest thermal conductivity at room temperature of any known material at $2200Wm^{-1}k^{-1}$ [1], high thermal conductivity allows for effective heat dissipation in turn allowing for higher power operation. Diamond is also mechanically tough and chemically inert [2]. Exhibits a high predicted breakdown field at $5 - 10MVcm^{-1}$ [1] allowing for high voltage applications. Diamond has the highest optical photon energy at $E = 160meV$ [1] of any semiconductor. High Optical photon energies can provide high carrier velocity saturation which is advantageous when fabricating high frequency electronics. A full comparison between diamonds electrical properties compared to those of other common semiconductor materials can be viewed in [Appendix 1].

However diamond electronics also have hurdles to overcome. Natural diamond has too many impurities for use in electronics so artificially manufactured diamond using CVD must be used. Recent developments in the field of synthesising diamond have allowed this hurdle to be overcome. Diamond's high band gap results in low activation at room temperature and most dopants available have even higher ionization energies [1]. Devices in which holes are the majority carriers are the only option available at the present time due to the absence of any convincing n-type dopants. The mechanical toughness and chemical inertness of the material also result in difficulties using conventional device fabrication, meaning unique fabrication techniques need to be developed.

Diamond in its intrinsic form is considered an insulator as it has a very large band gap (5.47eV), however when it is hydrogen terminated it exhibits high surface conductivity as discovered by Ravi and Landstrass [4], making it a viable semiconductor. This high surface conductivity is noted to be dependent on atmospheric adsorbates which act as acceptor donors through surface transfer doping as shown by Maier et al [5]. However these atmospheric adsorbates are unstable. An alternative method is to use another material in a surface transfer doping technique to provide surface acceptors to the hydrogen terminated diamond which are more stable. Some materials which have been explored previously are C60 and C60F46 [3] however it was demonstrated that these surface acceptors lack thermal stability and would certainly require passivation and encapsulation [3]. Another alternative material currently being explored is Molybdenum Trioxide (MoO3), a transition metal oxide. MoO3 is an attractive surface dopant for diamond as it has a high work function (higher than hydrogen terminated diamond thus providing a negative electron affinity and making electron transfer energetically favourable as demonstrated by Russell et al. [6]). It has also has previously been used to enhance electron/hole injection in organic electronics [6].

MoO3 has since been shown to perform suitable surface doping on semiconductor devices as demonstrated by Russell et al. [6]. However the devices have also shown a puzzling performance decrease over time. MoO3 is not a well characterised material. It is thought to be hygroscopic. Thin films of MoO3 have been shown to increase in thickness by around 1.5% over a period of 40 minutes exposure to the atmosphere in experimentation by Edward S. R. Bovill et al. [7]. This could be due to its proposed hygroscopic nature leading to a swelling of the film as molecules from the atmosphere diffuse into the bulk. However it has also been previously shown by Irfan et al. that the device performance degradation could be

attributed to oxidation as similar decreases to device performance were noted after exposure of thin films of the material to pure oxygen [8].

If the material is in fact hygroscopic it could attract and hold water through either adsorption or absorption. Adsorption being the adhesion of molecules or atoms to the surface of the film, which is proportional to the surface area of the film. Adherence of adsorbates to the surface of the film would be reversible by either a physical or thermal method. Absorption being the diffusion of atoms or molecules into the body of the film, which is proportional to the volume of the film. This possible absorption could decrease the effectiveness of MoO3 as a surface dopant as the particles suspended between the MoO3 molecules inhibit the transfer of acceptors to the diamond surface.

It has also been suggested that thermal evaporation of MoO3 can cause an oxygen deficient layer as noted by Irfan et al. [8], which could cause chemisorption of oxygen to the surface of the film. Bonds formed through chemisorption are stronger chemical bonds than those formed by Van der walls force of adsorption.

If absorption, adhesion or chemisorption occur a method of encapsulation or pacification must be developed to limit the degradation of MoO3 as a surface dopant.

Quartz crystal microbalance (QCM) is a technique whereby the mass per unit area of a quartz crystal resonator is monitored by measuring the change in its frequency as mass is added to the crystal's surface. The Sauerbrey Equations can be used to calculate the change in mass of the quartz crystal after mass has been deposited on it provided that: the deposited mass is rigid; the deposited mass is distributed evenly; and the change in frequency of the crystal is less than 2% [9]. Provided that the material is in fact hygroscopic a change in mass should be observed when the thin MoO3 films on resonant crystals are exposed to the environment. This mass increase could be caused by absorption of moisture, adsorption of moisture or chemisorption of oxygen.

The aim of this body of work is to monitor the possibility of absorption and adsorption of moisture, and the process by which oxygen affects MoO3 when exposed to atmosphere. This was attempted by verifying the thickness of films of MoO3 deposited using the MEB400 thermal evaporator and monitoring the mass of deposited thin films of MoO3 using quartz crystal microbalance before, during and after exposure to the environment.

Diamond Semiconductors

Impurity free Single crystal CVD diamond is in its intrinsic form almost considered an insulator with its high band gap of 5.5eV. When hydrogen terminated at the surface the energy band diagram shows a negative electron affinity (NEA) as the hydrogen's vacuum level lies below the diamonds conduction band as can be seen in *figure [1]*. This results in high surface conductivity as noted by Landstrass et al [4]. Electrons near the surface of the diamond readily ionise looking to leave the diamond if there is a suitable acceptor in contact with the diamonds surface. Although it is widely accepted that atmospheric adsorbates are the electron acceptors resulting in high surface conductivity in hydrogen terminated diamond it is currently not clear how this occurs. As Discussed by Stephen Russell [13] electron affinities of surface adsorbates are too low at around 2.3eV to accept electrons from the diamond, the acceptor must align properly with the lowest unoccupied molecular orbit at 4.2eV. However this can possibly be explained by the introduction of an aqueous layer and electrochemical transfer of charge as suggested by Maier et al. [5] as can be seen in *figure [2]*. The chemical potential of this aqueous layer has been calculated to be be in the region of -4.2 eV, meaning it would allow electrons to transfer from the diamonds surface leaving behind vacancies.

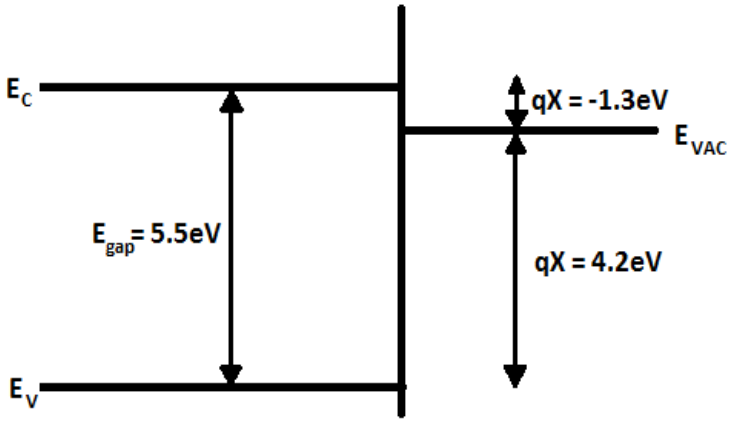


Figure [1] – Band diagram structure for energy of interface between diamond (left) and hydrogen termination (right).

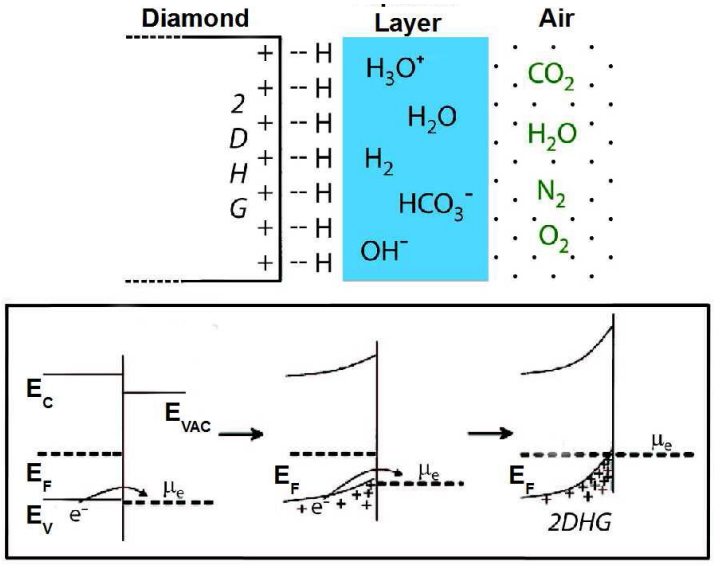


Figure [2] – Diagram showing aqueous region at hydrogen terminated diamond surface and transfer of charge resulting in electron deficient channel forming a 2 dimensional hole gas (2DHG) below the diamonds surface. Taken from "High Performance Hydrogen-terminated diamond field effect transistors, Stephen A. O Russell 2013"

These atmospheric adsorbates are however not stable, sublimating from the diamond surface at temperatures above 200 degrees. Thus an alternative method of providing surface acceptors to the diamond must be investigated. Alternative materials must have sufficiently high electron affinities to allow for favourable energy transfer from the diamond to the surface acceptor, in this case at least above 4.2eV. Some of the materials which have been previously investigated are fullerenes such as C60 and C60F48. These materials have shown good progress as acceptors but are arguably not more stable than atmospheric adsorbates sublimating from the diamond surface at around 250 degrees Celsius. From this information the decision to investigate inorganic materials as surface acceptors was made as they could prove to provide greater stability. As an alternative to surface adsorbates the material currently being investigated is MoO3. MoO3 provides high electron affinity and is more stable at higher temperatures than the organic compounds explored.

Absence of a diamond oxide means that another technique for device fabrication must be used as opposed to the traditional MOSFET. Metal Semiconductor Field Effect Transistors (MESFET) devices may provide an attractive alternative. A MESFET device could be designed to provide a bias at the gate to produce a depletion area allowing for charge from the source of the MESFET to flow to the drain utilising the sub surface 2DHG.

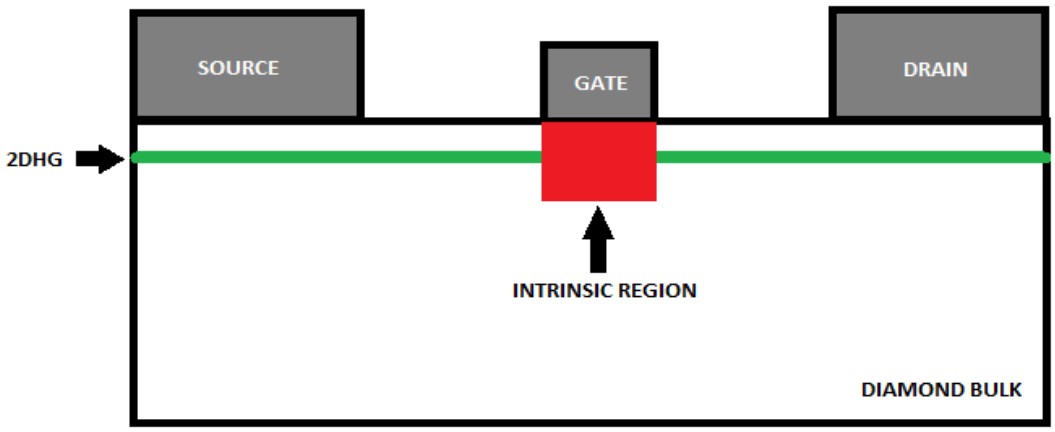


Figure [3]. Demonstrates the operation of a simple FET constructed using a surface doping technique to provide a 2DHG to form under the diamond substrates surface. Under the gate remains intrinsic diamond.

The interface between the diamond and the gate metal in the intrinsic diamond provides a built in potential between the gate and bulk, resulting in a depletion region dependent on the size of the gate. Due to the nature of p-type doping, a negative bias provides accumulation of holes at the surface in turn forming a channel with the 2DHG allowing current to flow between the source and drain. A positive bias will deplete holes away from the surface eventually leading to no current flow from source to drain. The Ohmic contacts of the source and drain allow current to flow through the 2DHG between source and drain provided that a negative bias is acting on the gate.

2. Theory

All experimentation carried out in JWNC clean room facility

Temperature = 296K

Humidity = 45%

2.1

Constants:

1 Angstrom = 1x10-10m

Density of water = 1000 g/cm^3

Boltzmann's constant =1.38x10-23

2.2

Fundamental properties of quartz crystal monitor:

Density of quartz = 2.648 g/cm^3

Shear modulus = $2.947x10^{11}$ g/cm^2

Area of crystal = 1.53 cm^2

Resonant Frequency of QCM = 6MHz

2.3

The Sauerbrey Equation is a method developed by Gunter Sauerbrey in the late 50's which correlates the change in frequency of a quartz crystal to additional mass added to it. The additional mass added to the face is treated as an extension of the crystal [9].

$$\Delta f = \frac{2f_0^2}{A\sqrt{\rho_q \mu_q}} \Delta m$$

$$\therefore$$

$$\Delta m = \frac{\Delta f . A\sqrt{\rho_q \mu_q}}{2f_0^2}$$

f_0 – Resonant frequency of crystal (Hz)
Δf – Frequency change (Hz)
Δm – Mass change (g)
A – Crystal area (cm^2)
ρ_q – Density of quartz ($g/cm^{3)}$
μ_q – Shear Modulus of quartz for AT-cut crystal (g/cm^2)

The formula can further be reduced by substituting the fundamental properties of the quartz crystal into a "Linear sensitivity factor": C_f. [9]

Thus

$$\Delta m = \frac{\Delta f}{C_f}$$

Where:

$$C_f = \frac{2f_0^2}{A\sqrt{\rho_q \mu_q}}$$

For this crystal:

$$= \frac{2 \times 6000000^2}{1.53\sqrt{2.648 \times 2.947\text{x}10^{11}}}$$

$$= 53271142\ Hz/g$$

2.4

Diffusion of oxygen into silicon can be given in the general form of mass diffusivity:

$$D_{OXY} = D_0 exp\left(\frac{-E_A}{k_B T}\right) cm^2 s^{-1}$$

D_{OXY} = Oxygen Diffusion coefficient in silicon ($cm^2 s^{-1}$)

D_0 = Maximum diffusion coefficient at infinite temperature ($cm^2 s^{-1}$)

E_A = Activation energy for diffusion (J)

k_B = Boltzmann's constant (JK^{-1})

T = Temperature (K)

In single crystal Czochralski silicon variables have been measured leading to [1]:

$$D_{OXY} = 0.13 exp\left(\frac{-2.53eV}{k_B T}\right) cm^2 s^{-1}$$

2.5

Volumetric Mass density can be given by:

$$\rho = \frac{m}{V}$$

ρ = Volumetric Mass density (Kgm^{-1})
m = Mass (Kg)
V= Volume (m^3)

3. Method of Experimentation:

All experimentation carried out in the James Watt Nanofabrication facility which is a controlled environment with humidity of 45% and temperature of 23 degrees Celsius.

3.1 MoO3 Deposition and characterisation:

Aim: Verify deposited thickness of film and characterise mechanically.

MoO3 was deposited onto clean silicon substrates using a thermal evaporation method. In this case using the MEB400 manufactured by "Plassys" in the James Watt Nanofabrication facility. Deposition occurred at a base pressure of $2x10^{-6}$ Mbar. The MoO3 was evaporated from pellets placed in a tungsten crucible onto a clean mechanical grade silicon substrate. In appendix [2] a full recipe for depositing MoO3 using the Plassys 5 has been provided. After deposition the Dektak stylus profiler was used to verify the thickness of deposited MoO3 on the substrates. To get an accurate measurement there must be a difference in height between the substrate surface and deposited films. To achieve this reference an additional piece of clean silicon will be clamped to the substrate to mask part of it from deposition. This was used as a reference to measure the difference in height of the deposited film and the substrate.. The samples were then mechanically characterised after the thickness of the sample was verified. Mechanical characterisation took the form of optical microscopy imaging and Scanning Electron microscope imaging (SEM) to verify that the sample was completely coated and the integrity of the deposition.

Silicon substrates coated with thicknesses of 40nm, 60nm, 80nm and 100nm of MoO3 and verified integrity and thickness. The Dektak was proposed to be used to monitor the thickness of the films as they are exposed to the environment. However due to the nature and accuracy of the stylus profiler meaning full data could not be achieved.

3.2 Quartz Crystal microbalance monitoring possible absorption and adsorption:

Aim: Monitor effects on thin films of MoO3 as exposed to the environment.

A clean resonant quartz crystal acquired from "Megatech Ltd." was installed into the Plassys 5 thermal evaporator. The crystal was placed under vacuum at $2x10^{-6}$Mbar as it would be during a deposition run. Next the chamber was vented to atmospheric pressure using nitrogen. At atmospheric pressure the chamber door was opened exposing the crystal to the environment. The resonating frequency of the crystal was monitored throughout. After exposure the crystals frequency was allowed to saturate and the chamber placed back under vacuum.

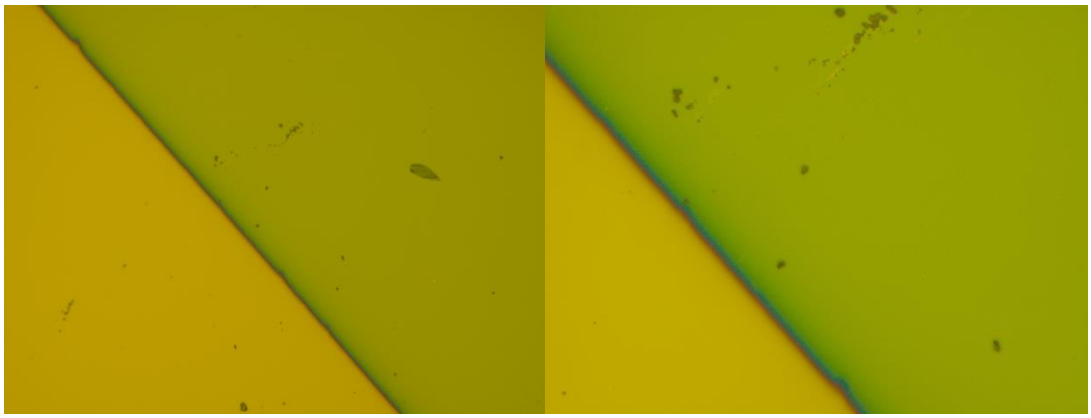
A clean quartz crystal was deposited with a 10nm thin film of MoO3. The frequency of the crystal was monitored as the crystal is vented from under vacuum at $2x10^{-6}$ Mbar back to atmospheric pressure. The chamber door was then opened and the crystal left in the atmosphere until the frequency response of the crystal saturated. Next the crystal was placed back under vacuum for direct comparison of frequency under vacuum before and after the film was exposed to the environment.

A clean quartz crystal was deposited with a 50nm thin film of MoO3. The frequency of the crystal was monitored as the crystal is vented from under vacuum at $2x10^{-6}$ Mbar back to atmospheric pressure. The chamber door was then opened and the crystal left in the atmosphere until the frequency response of the crystal saturated. Next the crystal was placed

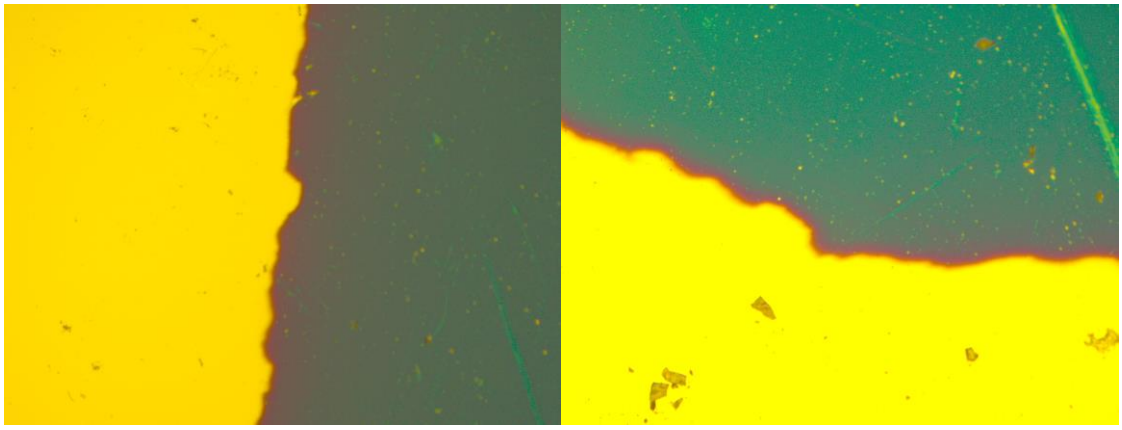
back under vacuum for direct comparison of frequency under vacuum before and after the film was exposed to the environment.

4. Results

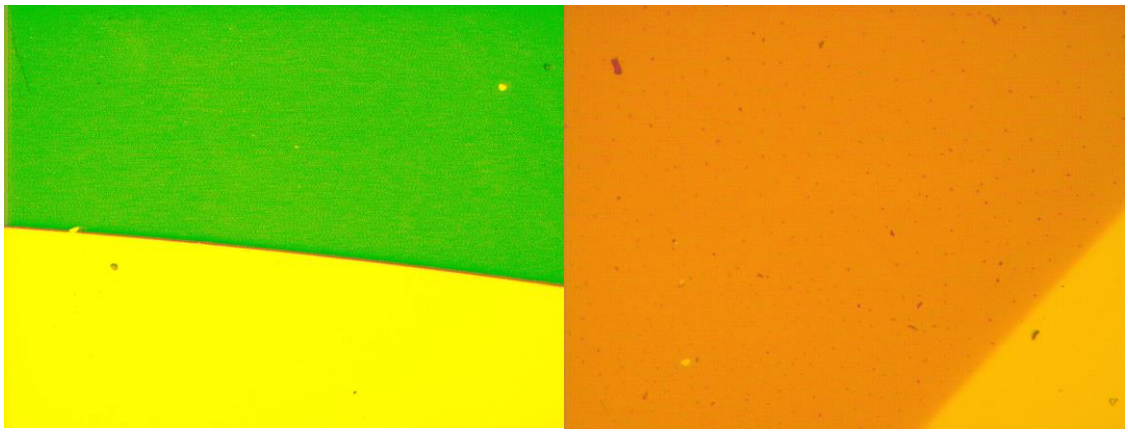
4.1 – MoO3 Deposition and Characterisation



[Figure 4.1] Two images of 100nm MoO3 deposited on clean silicon substrates films showing shielded transition from MoO3 to silicon substrate. Samples masked using an additional piece of silicon clamped to its surface.



[Figure 4.2]. Two images of 60nm MoO3 deposited onto silicon substrates shielded using a clamp.



[Figure 4.3] 80nm sample (left) 40nm sample (right) of MoO3 both masked using silicon clamped to the substrates surface.

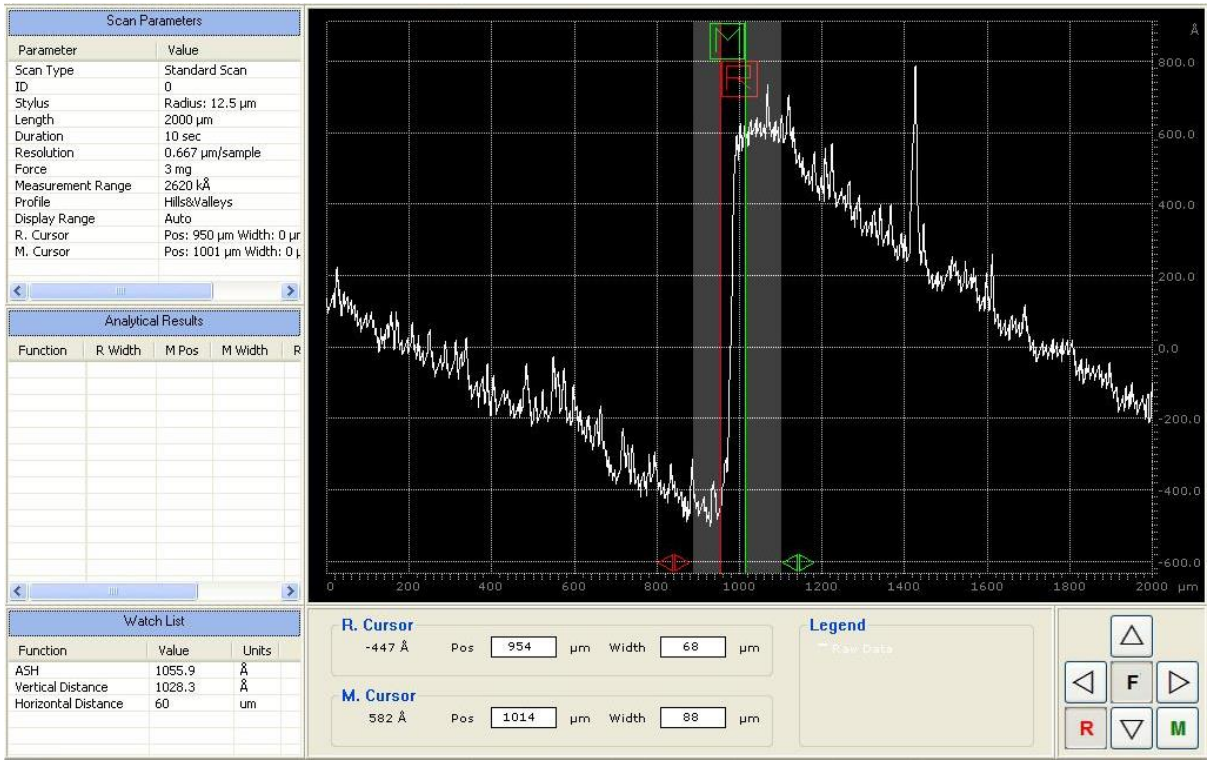
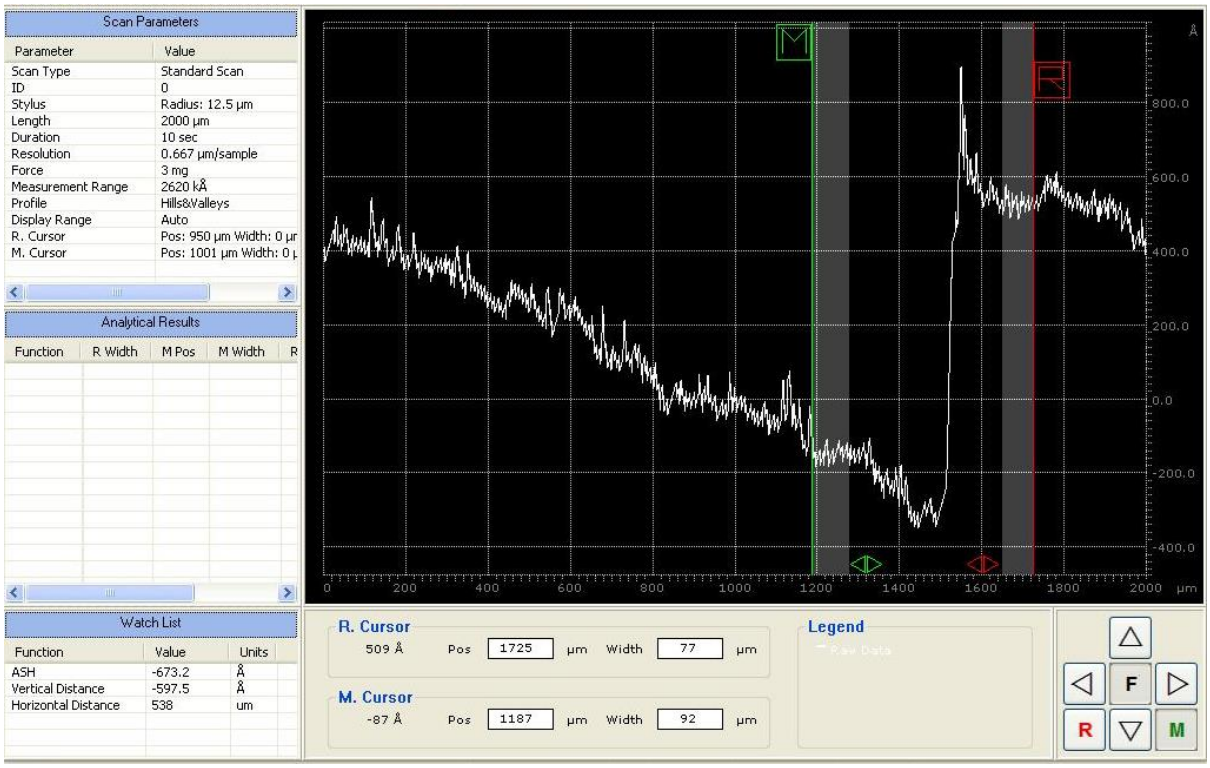
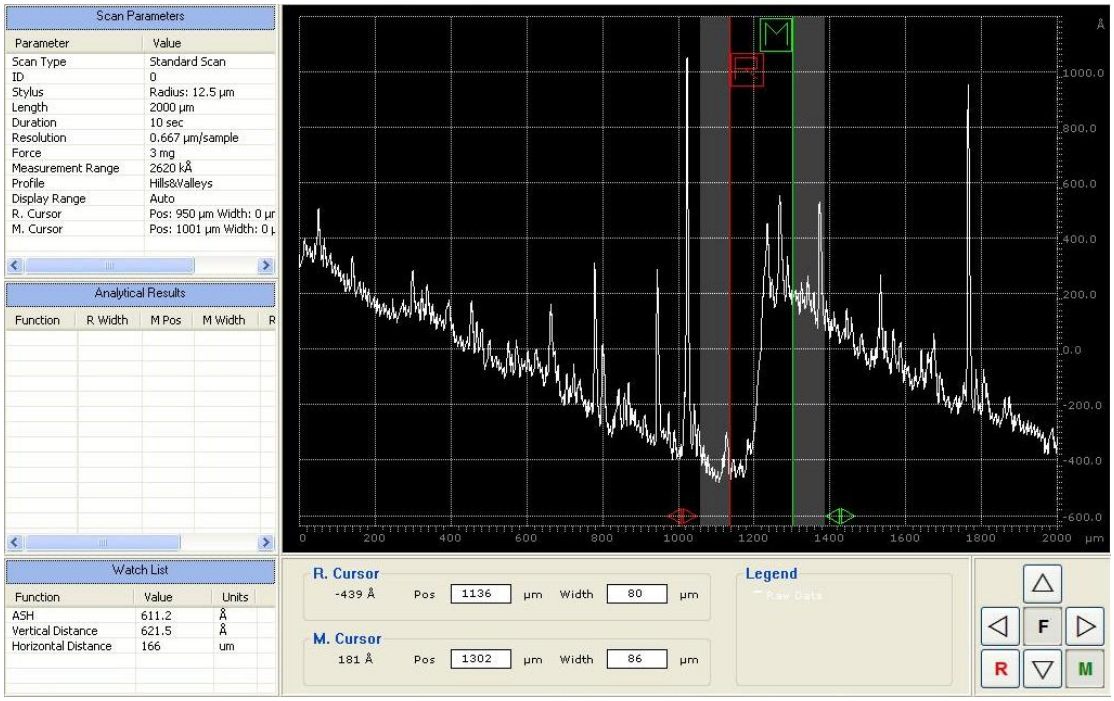


Figure [4.4] Screenshot of height verification using Dektak stlus profiler of 100nm MoO3 deposited film immediately after deposition. Note the majority consistency of the surfaces.



[Figure 4.5] Screenshot of height verification using Dektak stlus profiler of 60nm MoO3 deposited film immediately after deposition



[Figure 4.6] Screenshot of height verification using Dektak stlus profiler of 60nm MoO3 deposited film after exposure. Note the large spikes which are assumed to be particles resting on the surface.

4.2 – MoO3 film exposure

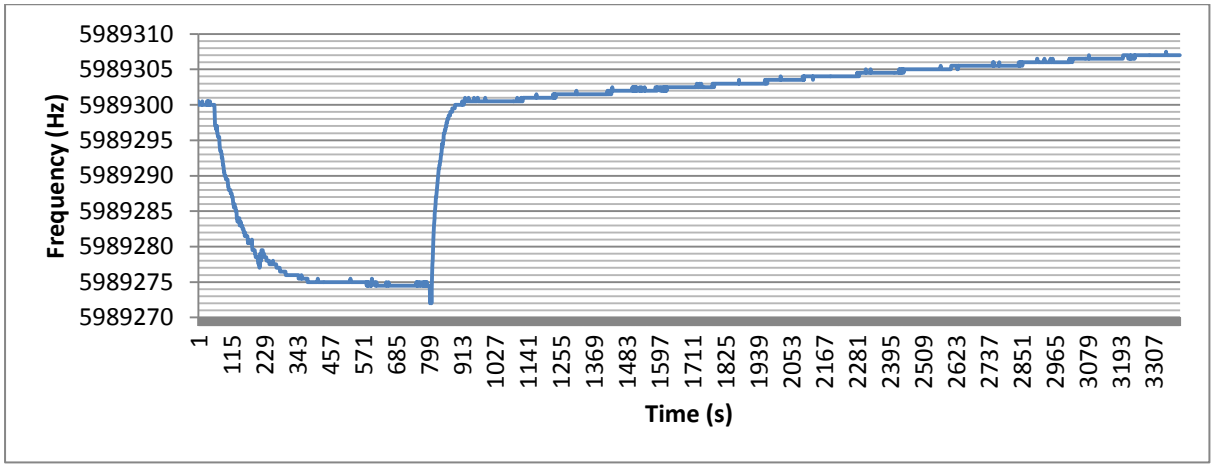


Figure [4.7] Clean Quartz Monitor Crystal frequency response during exposure to environment and reapplication of vacuum.

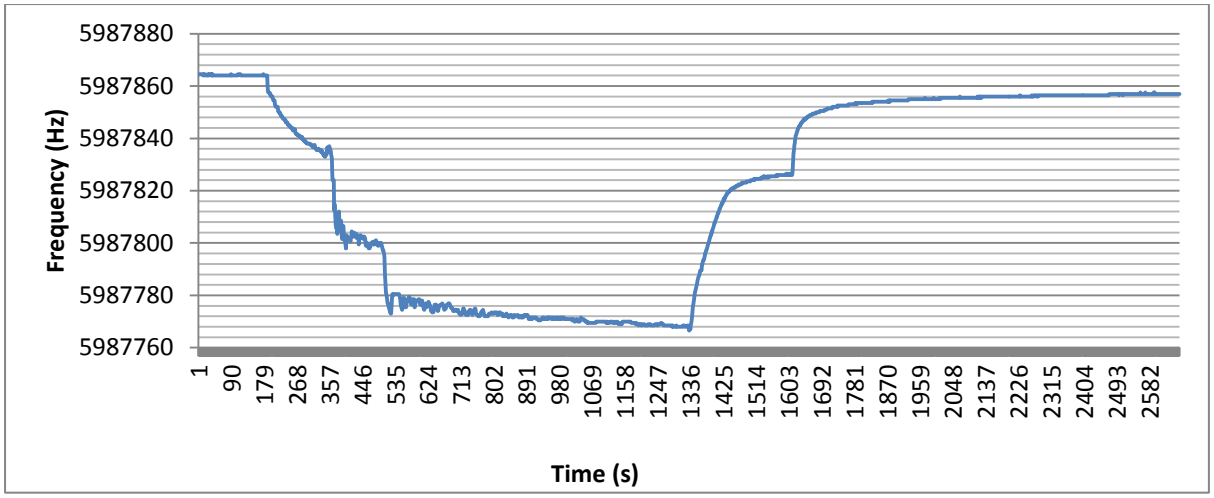


Figure [4.8] 10nm deposited thin film of MoO3 on clean crystal frequency response after exposure and reapplication of vacuum.

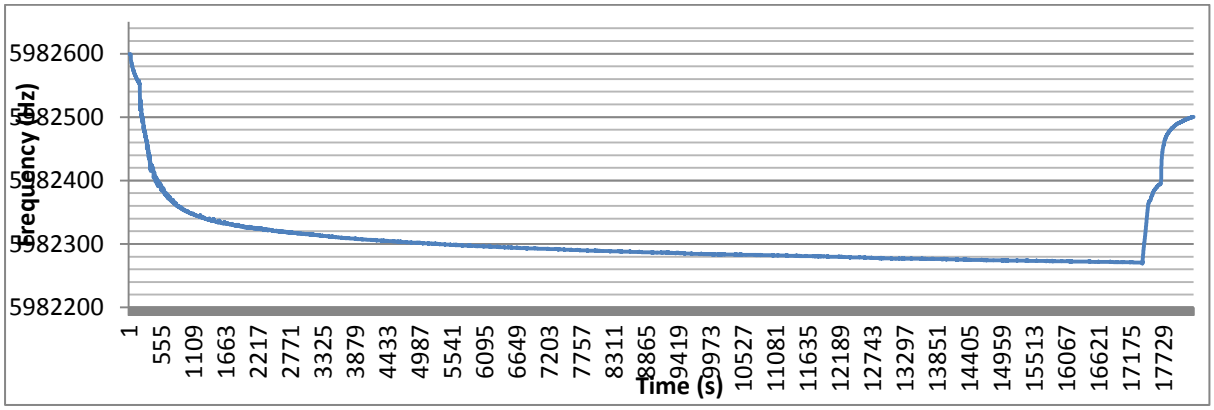


Figure [4.9] 50nm deposited thin film of MoO3 frequency response after exposure and reapplication of vacuum.

5. Discussion

5.1 – MoO3 deposition and Characterisation

Target Thickness (Å)	Measured Thickness (Å)	Error (%)
400	411	2.75
600	621	3.50
800	819	2.38
1000	1028	2.80

Figure 5.1.1: Thickness of MoO3 deposited on clean silicon substrates. Each thickness was attempted twice and multiple Dektak traces were taken.

As can be seen from the data gathered the deposition of MoO3 by thermal evaporation is very accurate (within 30 Angstroms for most depositions. This method of thermal evaporation claims to be within 1nm (10 angstroms), the inaccuracy in these measurements compared to the specified accuracy could be attributed to the method of measurement. The accuracy of deposition appears to lie on average around 3% of the desired thickness. When dealing with such small dimensions accuracy within 3% is excellent.

Optical microscopy and Scanning Electron Microscopy (SEM) were used to review the integrity of deposited MoO3 on clean silicon substrates. Mechanically the deposition of the material appears even at each thickness. On clean substrates it can be expected to receive an even coating of MoO3 across the exposed sample face. At the shielded edges of the deposited MoO3 where it transitions from MoO3 to the silicon substrate the coating can be sudden or have a more smooth transition depending on the quality of shielding. It was found that using an additional piece of silicon clamped as a shield provides a better mask than using only the area covered by the clamp to monitor the transition.

Each thickness of deposition appears to produce a slightly different colour, I would expect this is due to the refraction from different thickness of MoO3 as light reflects off the clean silicon underneath. This would imply that MoO3 at very thin films is semi-transparent. This may be interesting to further investigate as if MoO3 is semi-transparent it would allow photons to pass through it. The material could be utilised for optical electronics.

Attempts to verify the thickness of the samples with time from exposure to the environment were not successful due to the nature of the Dektak stylus profiler. It is almost impossible at this scale to measure the exact area of the sample twice. An alternative method being explored was to use the AFM in the JWNC, however time constraints did not allow for this experimentation to be completed. The values for multiple Dektak traces were deemed to be of little use for measuring any increased volume of the films deposited.

Had the experimentation worked successfully It would have been expected to see swelling of the film, indicative of increased volume from absorption of moisture. As noted by

Edward S. R. Bovill et al. whom observed an increase in thickness of MoO3 films when exposed to the environment over a 40 minute period [4].

5.2 – 10nm MoO3 deposition exposure

When a clean quartz monitor crystal is exposed to the environment it shows a total decrease in frequency of around 25Hz. This can be attributed to the resonation of the crystal being damped as the medium changes from under vacuum to being surrounded by a gas. From the graph it can be seen that around 80% of the decrease can be attributed to the nitrogen venting into the chamber. The bump which can be seen after 20Hz decrease in frequency is the door of the chamber opening after reaching room pressure (1013 mBar at sea level [10]) After being vented with nitrogen the door of the chamber is opened allowing air to enter. As the atmosphere is 78% nitrogen in volume [10] this can be expected as the density of the surrounding gas will change little between nitrogen and air. The frequency of the crystal saturates very quickly once exposed to the environment (around 3 minutes of exposure). This can be expected as there is no chemical change and saturation of moisture and particles settling on the surface can be expected to happen very quickly on a small area after exposure to the atmosphere. When the crystal is placed back under vacuum it was noted that the frequency returns to its original value after the rough vacuum pump to a pressure of 1x10-2 Mbar and increases linearly until it returns to its original vacuum pressure where the frequency can be seen to have risen by around 5Hz above its original value. This experiment was carried out 4 times and the same frequency decrease was found each time.

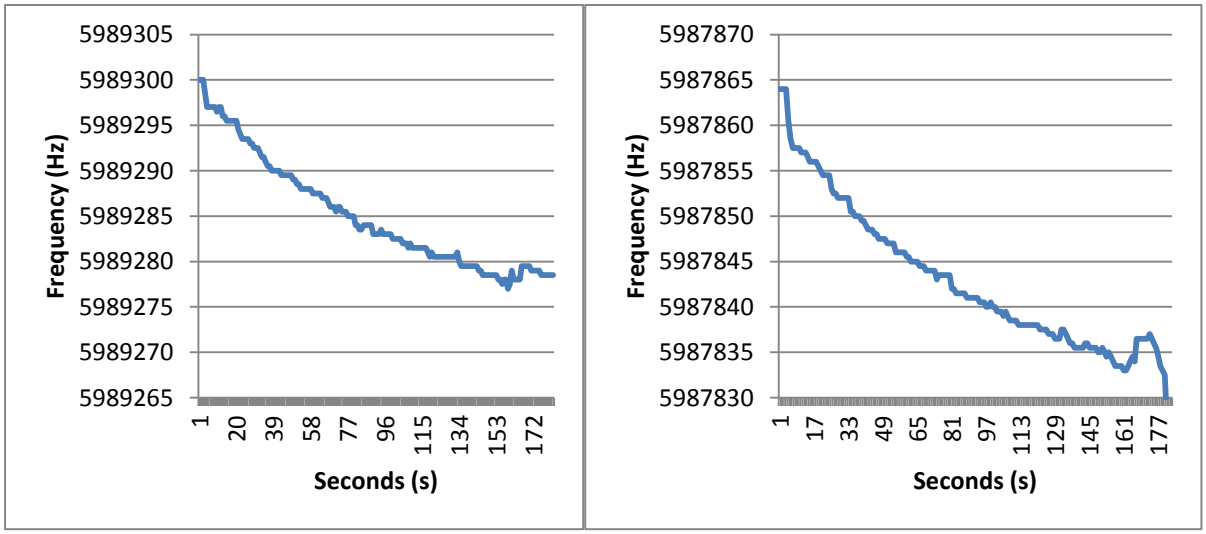


Figure [5.2.1]: Comparison of nitrogen venting for clean crystal (Left) and crystal deposited with 10nm MoO3 (Right).

After 10nm deposition has been performed on a clean quartz crystal the frequency is allowed to saturate under vacuum. Next the chamber vents with nitrogen in around the same time as fresh crystal as can be seen from the graphs (~150 seconds to vent to room pressure) However whereas the nitrogen venting fresh crystal can be seen as reducing by <25 Hz the 10nm sample can be seen to decrease by around 30 Hz before exposure to the atmosphere. This could be explained by the slight increase in surface area as the 3 dimensional film is deposited on the crystal. Immediately after exposure to the environment it can be seen that

there is a stark difference between the 10nm deposition and clean crystal. The 10nm sample suddenly begins to decrease in frequency rapidly until it plateaus, and then decrease rapidly again in an exponential fashion until the frequency was seen to saturate after 20 minutes of exposure. After this the chamber was put back under vacuum. This large decrease in frequency in contrast to the small change on a fresh crystal is clear evidence of an increase in mass of the film as it is exposed to the environment. It is not clear what process caused the plateau after the initial exposure to the environment; I would have expected to see an exponential decay from exposure to saturation. Perhaps this could be caused by some deposition error, such as a large quantity of MoO3 becoming stuck together. Another explanation could be the saturation of mass change associated with adsorption to the surface of the film happening earlier than diffusion of moisture into the bulk beginning. The saturation of adsorption could certainly be expected to happen much faster than the saturation of absorption as absorption is dependent on the volume of the films as opposed to the surface. A repeat of the experimentation would be beneficial. The 10nm sample saturates after the frequency decreases by a total of 95Hz. We can assume that the volume of the monitor crystal has increased by a negligible amount, and thus can subtract the frequency decrease from medium change associated with the clean crystal experiment 25Hz, providing a total decrease with atmospheric exposure of 70Hz.

$$\Delta m = \frac{\Delta f}{-C_f}$$

$$\Delta m = \frac{70}{C_f}$$

$$\Delta m = 1.314 \times 10^{-6} g$$

This change in mass with exposure to the environment can be attributed to a combination of absorption and adsorption. It can be noted that as the increase in mass has saturated this would imply the material is saturated with moisture diffused into the bulk, as well as a saturation of moisture adhering to the surface of the film.

It has been noted that thermal evaporation of MoO3 can cause an oxygen deficient film, thus oxygen could be absorbed into the substrate. I propose that some of the additional mass is caused by chemisorption of oxygen by the oxygen deficient film [8]. XPS or Auger spectroscopy could be used to monitor the chemistry after atmospheric exposure, thus it could be conclusively shown what has been absorbed by the film.

After Reapplication of the vacuum the frequency can be seen to increase back to around 15Hz below its initial value. If we factor in the increase after reapplication of vacuum on the clean crystal we can take this as a permanent increase in mass associated with a permanent decrease in the crystal resonation of 20 Hz.

Using Sauerbrey equation the permanent change in mass associated with this films absorption is:

$$\Delta m = \frac{20}{C_f}$$

$$\Delta m = 3.754 \times 10^{-7} g$$

This mass change of the film with exposure can be attributed to absorption. It can be noted that moisture adhering to the surface of the MoO3 film should be removed under vacuum leaving a permanent mass change of moisture diffused into the film.

As we know the volume and density of the deposited MoO3 we can calculate the theoretical mass of the film.

$$m = \rho . V$$

$$m = 4.46 \times (1.53 \times 10 \times 10^{-7})$$

$$m = 6.823 \times 10^{-6}$$

From this we can note that the irreversible increase in mass is 5.5% of the total mass. This is clear evidence of a permanent change with atmospheric exposure caused by MoO3's hygroscopic nature.

5.3 – 50nm MoO3 deposition exposure

The 50nm sample decreases by 45 Hz with venting of nitrogen. This larger decrease associated with the nitrogen venting may again be attributed to the increased surface area from depositing material onto the crystal face. This seems plausible as there has been a larger increase in the frequency with the larger surface area attributed to the 50nm sample.

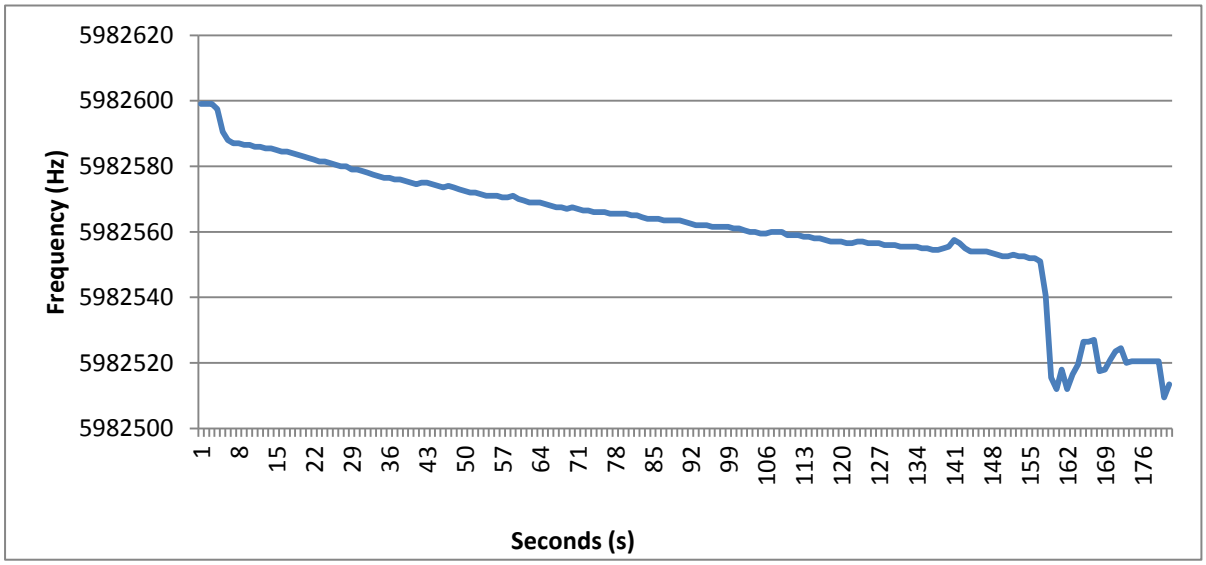


Figure [5.3.1]: Nitrogen venting of crystal with 50nm of MoO3 deposited.

After the door of the chamber is opened at atmospheric pressure the frequency of the crystal decreases rapidly in an exponential fashion, the crystal was left to fully saturate which took a total of over 5 hours of exposure. The crystal frequency saturates after a decrease of 325Hz, subtracting 25 Hz associated with change in medium we get 300Hz change at saturation for a 50nm sample. This results in a mass change of:

$$\Delta m = \frac{300}{C_f}$$

$$\Delta m = 5.631 \times 10^{-6} g$$

It appears the relationship of time until saturation and thickness of deposition is not linear. This could be indicative of the rate of absorption decreasing as the substance becomes more inundated with moisture. This can be expected as the greater thickness of MoO3's bulk should take a substantially longer time than a smaller bulk to saturate with absorption of moisture. In this case it can be seen that the saturation of 50nm takes around 15 times longer than 10nm.

If we take the average rate of change of both the 10nm and 50nm depositions before they become linear we can see that the 10nm sample has a rate of change of 0.136 and the 50nm; 0.14. This can be expected as we would expect the 50nm sample to have a negligibly larger surface area than the 10nm. Thus as absorption occurs through the surface into the film the rates at which absorption occurs should be close; with the 50nm slightly more. As time goes on however the rate of change of the 10nm should be less than the 50nm sample as the 10nm sample saturates more quickly with its smaller volume.

After the saturation of frequency decrease caused by increased mass of the film the vacuum was reapplied to the chamber. When the chamber was pumped back to its original post deposition pressure the frequency of the crystal was 100 Hz below its original value. Factoring in the frequency increase from the clean crystal experiment this leaves us with a permanent mass change of 105 Hz. This is around 5 times greater than the 10nm sample as would be expected with a volume change of this order (volume is 5 times greater). This would be indicative of absorption as absorption is the diffusion of molecules or atoms into another substance, thus a larger volume would allow for more absorption and thus a larger mass increase in the film.

For 50nm using the Sauerbrey equation we get a permanent change in mass of:

$$\Delta m = \frac{105}{C_f}$$

$$\Delta m = 1.971 \times 10^{-6} g$$

As expected the permanent mass change is around 5 times greater than for the 10nm sample as the volume is 5 times greater. This would be indicative of absorption as the saturation of additional mass should be directly proportional to the volume of the bulk of the film. It can also be noted that the permanent change in mass is only one third of the total mass change. This implies that adsorption plays a larger part in the mass increase than absorption.

$$m = \rho . V$$

$$m = 4.46 \times (1.53 \times 50 \times 10^{-7})$$

$$m = 3.412 \times 10^{-5} g$$

It can be seen that this increase in mass is 5.7% of the total mass of the 50nm sample, In line with the 5.5% mass increase from the 10nm sample.

If we take into account the density of water (1g/cm3) we can calculate the volume this mass of water should be. In comparison to the overall volume of the film we can gather how much the film has swollen with this absorption.

$$V = \frac{m}{\rho}$$

$$V = \frac{1.971 \times 10^{-6}}{1}$$

$$V = 1.971 \times 10^{-6} cm^3$$

Compared to the theoretical volume of the film we could take this increased mass assuming it is water as an increase in volume in the region of 25% of the films volume. This is not what I would expect or what has been noted before in the literature. I would suspect this could be attributed to either erroneous calculation or limitations of the Sauerbrey equation [9]. It could also be attributed to the fact water is almost 5 times less dense than MoO3 and the additional mass would therefore cause a larger swelling. I would propose this is a sign that not all of the added mass is attributable to moisture absorption. As I would it has not been seen previously such a large increase as has been noted theoretically. I take this as a sign that there is more than just hygroscopy with thermally evaporated MoO3 exposed to the environment. I would also note that the moisture could in fact be absorbed into vacancies already in the MoO3 film after thermal evaporation. This would reduce the swelling of the film substantially. It should also be noted that putting the films back under vacuum may not result in the removal of all moisture from the surface of the film through adsorption. The

point pressure provided by the vacuum may not be enough to remove all the particles adhering to the surface through van der walls force

I would propose that the permanent change in mass is due to absorption of moisture and chemisorption of oxygen and the change in mass reversed by the reapplication of the vacuum is adhesion of moisture and other particles on the surface of the MoO3 film. The fact that the mass not under vacuum changes drastically with deposition of the film opposed to the clean crystal I feel this indicative of adhesion of moisture to the surface of the film possibly caused by adsorption at the surface of the film. The adsorption would possibly be reversed by the reapplication of the vacuum resulting in a reversal of frequency decrease as seen here. The permanent change could also be caused in part by chemisorption of oxygen in an oxygen deficient film as is sometimes caused by thermal evaporation of MoO3.

It has been suggested that the mass increase could also be from diffusion of oxygen into the substrate. However apart from chemisorption of oxygen I believe that it can be shown that the temperature and time scale needed to diffuse oxygen into metals is much higher than provided here, for instance 450 degrees Celsius in silicon is considered low temperature for oxidation. Which although not the same material as MoO3 it can be used as a reference to calculate a reasonable figure on times for oxygen diffusion as it is the best characterised material on the planet. From the formula for oxygen diffusion in silicon at low temperatures [11]:

$$D_{OXY} = 0.13\exp\frac{-2.54eV}{k_B T} cm^2 s^{-1}$$

$$= 0.13\exp\frac{-2.54 \times 1.6 \times 10^{-19}}{1.38 \times 10^{-23} \times 296} cm^2 s^{-1}$$

$$D_{OXY} = 1.19 \times 10^{-44} cm^2 s^{-1}$$

It can be seen that the diffusion coefficient is far too small to provide any large mass increase at these temperatures over the period of time the MoO3 film was exposed for. Even after 5 hours of exposure the oxygen would be expected to diffuse around 2x10^-40 cm^2. There is however a chance that oxygen could permeate the material and find itself in vacancies existing in the physical nature of the material. To investigate this it would be required to carry out XPS or Auger Analysis on the film to gather the chemicals which are present.

It can be seen that there is certainly a process at work resulting in the mass of thin films of MoO3 increasing with exposure to the environment. This is clear evidence that MoO3 is hygroscopic. I propose it is affected by both adsorption and absorption with adsorption providing a larger mass increase than absorption. A more in depth analysis may be able to conclude the part chemisorption plays in the increased mass.

Conclusion

From the experimentation it is clear to see there is some process at work resulting in an increased mass in deposited films of MoO_3 when exposed to the environment. It can also be noted that adsorption and absorption each play a part in the increased mass of the films. When the exposed films are put back under vacuum we see a reversal in some of the mass increase of the film, this is indicative of adsorption of moisture adhering to the surface of the film by weak van der walls force being removed at the surface of the film. Comparing the mass change under vacuum and at room atmosphere it can be seen that the mass change under return to vacuum is around a third of the total mass change. This shows that the mass of moisture on the surface through adsorption is larger than the mass of moisture absorbed into vacancies in the bulk of the film. However absorption may not be the only permanent change. It has been shown that thermal evaporation of MoO_3 can cause oxygen deficient films to be deposited [8], allowing for some chemisorption of oxygen into the film. This would also be a permanent change as the force of the vacuum would not be strong enough to break the newly formed chemical bonds.

The data gathered during experimentation shows that adsorption provides a larger increase in mass than absorption. However absorption and chemisorption are still the primary processes resulting in a decrease of the films attractive qualities as a dopant. This can be suggested as adsorption acting on the surface of the film should not have any effect on the concentration of acceptors available to be donated from the MoO_3 to the diamond substrate. The absorbed moisture into the bulk of the film however should decrease the number of acceptors available to the diamonds surface. It may also result in increased scattering of carriers within the diamond leading to decreased carrier mobility. Chemisorption may also introduce additional oxygen atoms into the substrate which could lead to a decrease in concentration of acceptors. However the moisture adsorbates on the surface could of course be absorbed into vacancies in the substrate over time until the MoO_3 film is saturated with moisture. Thus if molybdenum trioxide is to be utilised in the future of diamond semiconductor devices techniques of encapsulation and passivation will need to be developed to prevent adsorption, and absorption of moisture affecting the films as well as preventing chemisorption in oxygen deficient films.

Further Work

As a first step towards understanding MoO3 as it is exposed to the environment I would suggest repeating my own thickness characterisation using another method of monitoring possible swelling of the film. Spectral Ellipsometry could be an alternative method however it relies on calculations using the refractive index of a material which could change as the film is exposed. I would suggest a direct height measurement method such as Atomic Force Microscopy to gain an accurate picture of exposures effects on the thickness of thin films of MoO3.

The next step I would propose in the investigation of MoO3 hygroscopy would be to carry out X-ray Spectroscopy (XPS) or alternatively Auger Spectroscopy to monitor what chemicals are present in the film substrate after exposure to the environment. Both XPS and Auger spectroscopy use the energy of observed particles to determine their chemical composure as explained in Surface Analytics [12]. This could be used to conclusively answer whether the chemisorption of oxygen provides a substantial increase in mass of the films when compared to moisture absorption. This could in turn lead to a conclusion about whether absorption of moisture, chemisorption or diffusion of oxygen is the primary processes resulting in decreased performance of devices fabricated using MoO3.

Additionally I would suggest fabricating diamond devices with MoO3 as a surface acceptor to monitor the electrical characteristics of the devices as they are exposed to the environment. Van der Pauw structures could be fabricated on the material to determine the density of the majority carrier and the mobility of carriers in the device. These could be monitored as the un-encapsulated MoO3 is exposed to the environment to monitor if there is degradation. I would expect to see a decrease in both as absorption occurs.

Finally I would conclude this study by investigating methods of encapsulating MoO3 to prevent the processes by which its electrical characteristics are negatively affected. This could be done by depositing another material onto the MoO3 protecting it from exposure or perhaps by annealing the MoO3. Annealing was shown by Edward S. R. Bovil et al. to reduce the swelling of films of Molybdenum oxides [7] perhaps this would be enough to combat absorption. However the process by which annealing reduces swelling of the films is not clear at this time and may also need to be investigated.

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Appendix 1.

Properties of common semiconductor materials compared to Diamond.

Note: Figures of merit are normalised at SI = 1

	Si	4H-SiC	GaN	Natural Diamond	CVD Diamond	Potential device application benefit
Bandgap (eV)	1.1	3.2	3.44	5.47	5.47	High temperature
Breakdown field ($MVcm^{-1}$)	0.3	3	5	10	10	High voltage
Electron saturation velocity ($x10^7$ cm s^{-1})	0.86	3	2.5	2	2	High frequency
Hole saturation velocity ($x10^7$ cm s^{-1})	n/a	n/a	n/a	0.8	0.8	
Electron mobility ($cm^2 V^{-1} s^{-1}$)	1450	900	440	200–2800	4500	
Hole mobility ($cm^2 V^{-1} s^{-1}$)	480	120	200	1800–2100	3800	
Thermal conductivity ($Wcm^{-1} K^{-1}$)	1.5	5	1.3	22	24	High power
Johnson's figure of merit	1	410	280	8200	8200	Power-frequency product
Keyes' figure of merit	1	5.1	1.8	32	32	Transistor behavior thermal limit
Baliga's figure of merit	1	290	910	882	17200	Unipolar HF device performance

From "Diamond as an Electronic Material, Chris H. J. Wort and Richard S. Balmer, Element Six, Materials today, jan-feb 2008, Volume 11, Number 1-2,"

Appendix 2.

Sub Processes for MoO3 Deposition using MEB400

- Crucible MoO3
- Egun Emission Ramp – 40A, 01:00
- Egun Emission Ramp – 100A, 01:00
- Egun Emission Ramp – 120A, 01:00
- Wait 00:10
- Rate control 0.2nm/s
- Zero Thickness
- Shutter Open
- Wait for Termination
- Shutter Closed
- Egun Emission Ramp – 0.0A 00:30

Appendix 3 – Specification

2. Specification
Project Title - Making nano-patterns on transition metal oxides for diamond-based electronic devices.

Topics	Specification
Material Deposition:	- Characterise material visually and mechanically (Electron Microscope and AFM probe). - Should be able to accurately measure thickness of the deposition.
Calibrating temperature:	-How long does it take the Plassys MEB 400 substrate holder to get to 400C (operating temperature)? Use external measure of temperature (thermocouple for example) - Comparison with temperature from software readings of Plassys MEB 400.
Absorption:	- MoO3 is hydroscopic. How much does the mass of the film change after exposure to the environment? Can use the density of water to calculate how much moisture is absorbed. - In theory the mass should increase as the sample absorbs moisture from the environment. -resonant crystal within Plassys MEB 400 can possibly be used to observe change in mass. Noise may be an issue.
Exposure to E-Beam:	-Expose film to E- Beam -Check Mechanical and visual characterisation -Check Electrical Characterisation (Van der Pauw and Hall Measurements.) -Electrical Characterisation may be sensitive to electron beam.